



ADS EXP 7: Outlier detection using distance based/density based method

```
In [1]: # import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neighbors import NearestNeighbors
```

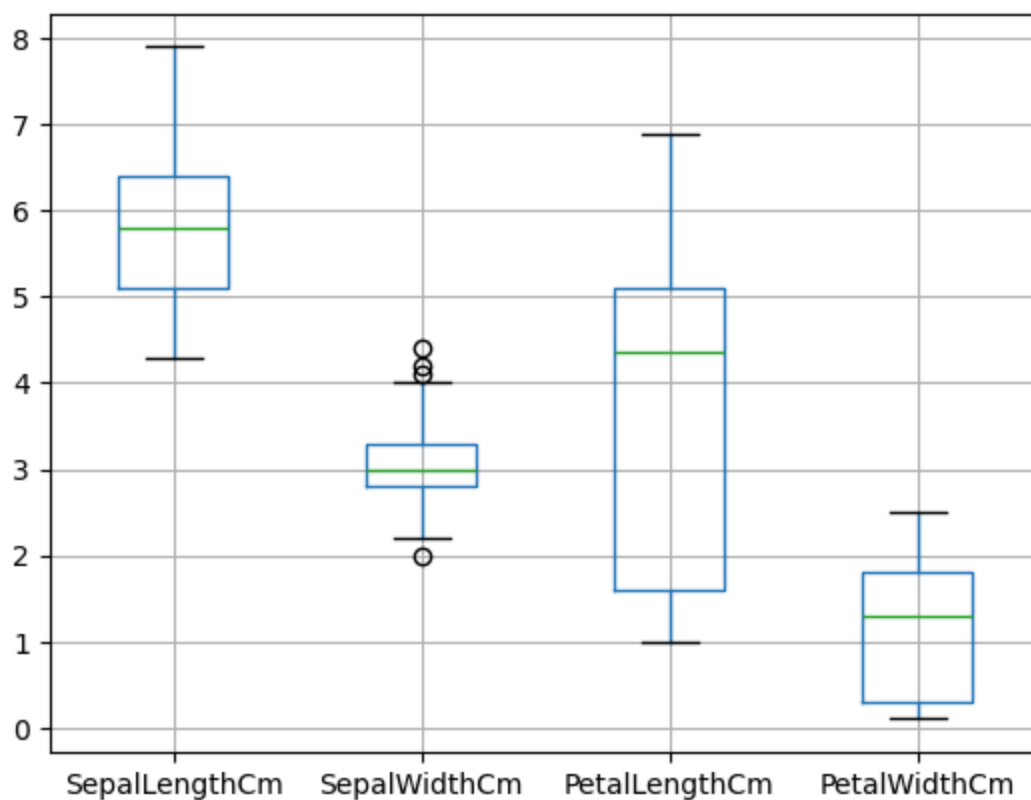
```
In [3]: data = pd.read_csv('Iris.csv')
data.head()
```

```
Out[3]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [4]: new_data=data.drop(['Id'],axis=1)
new_data.boxplot()
```

```
Out[4]: <Axes: >
```



```
In [5]: # create arrays
X = new_data.drop('Species',axis=1).values

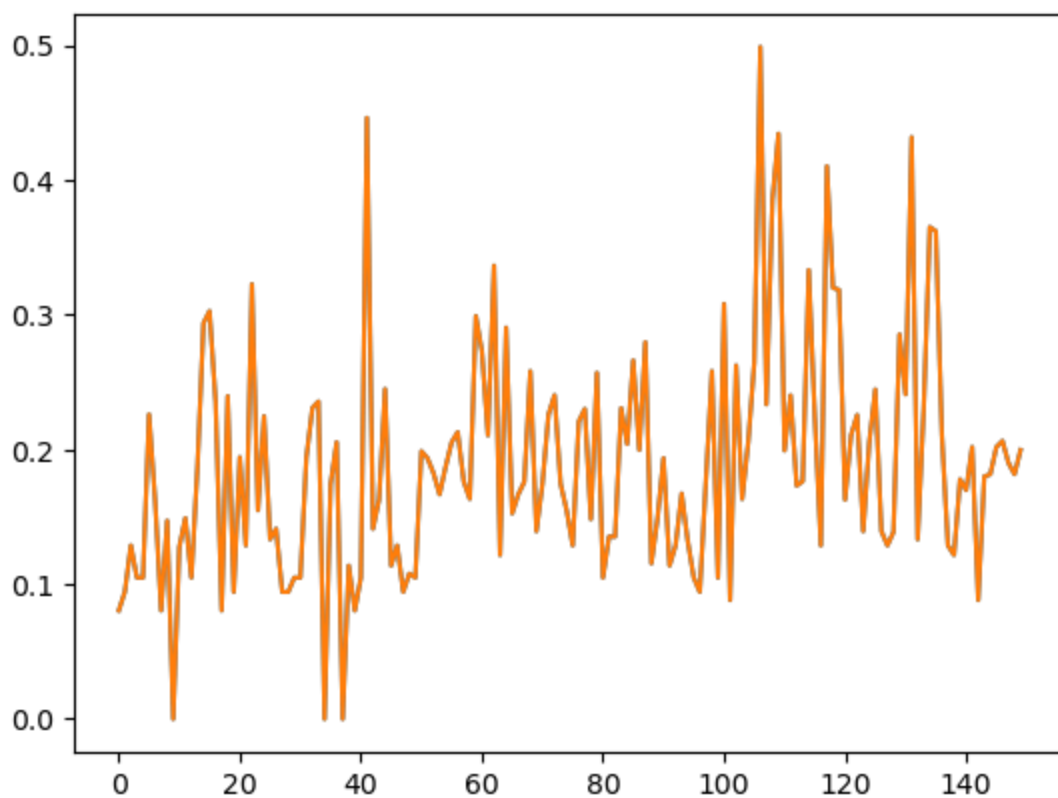
# instantiate model
nbrs = NearestNeighbors(n_neighbors = 3)

# fit model
nbrs.fit(X)
```

```
Out[5]: ▾ NearestNeighbors ⓘ ?
NearestNeighbors(n_neighbors=3)
```

```
In [7]: # distances and indexes of k-neighbors from model outputs
distances, indexes = nbrs.kneighbors(X)
# plot mean of k-distances of each observation

plt.plot(distances.mean(axis = 1))
plt.show()
```



```
In [8]: # visually determine cutoff values > 0.15
outlier_index = np.where(distances.mean(axis = 1) > 0.3)
outlier_index
```

```
Out[8]: (array([ 15,  22,  41,  62, 100, 106, 108, 109, 114, 117, 118, 119, 131,
                134, 135]),)
```

```
In [9]: # filter outlier values
outlier_values = new_data.iloc[outlier_index]
outlier_values
```

Out[9]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
15	5.7	4.4	1.5	0.4	Iris-setosa
22	4.6	3.6	1.0	0.2	Iris-setosa
41	4.5	2.3	1.3	0.3	Iris-setosa
62	6.0	2.2	4.0	1.0	Iris-versicolor
100	6.3	3.3	6.0	2.5	Iris-virginica
106	4.9	2.5	4.5	1.7	Iris-virginica
108	6.7	2.5	5.8	1.8	Iris-virginica
109	7.2	3.6	6.1	2.5	Iris-virginica
114	5.8	2.8	5.1	2.4	Iris-virginica
117	7.7	3.8	6.7	2.2	Iris-virginica
118	7.7	2.6	6.9	2.3	Iris-virginica
119	6.0	2.2	5.0	1.5	Iris-virginica
131	7.9	3.8	6.4	2.0	Iris-virginica
134	6.1	2.6	5.6	1.4	Iris-virginica
135	7.7	3.0	6.1	2.3	Iris-virginica

```
In [10]: # data wrangling
import pandas as pd
# visualization
import matplotlib.pyplot as plt
# algorithm
from sklearn.cluster import DBSCAN
```

```
In [11]: # input data
df = data[["SepalLengthCm", "SepalWidthCm"]]
# specify & fit model
model = DBSCAN(eps = 0.4, min_samples = 10).fit(df)
```

```
In [12]: # visualize outputs
        colors = model.labels_
        plt.scatter(df["SepalLengthCm"], df["SepalWidthCm"], c = colors)
```

Out[12]: <matplotlib.collections.PathCollection at 0x21fc44b4830>

```
In [13]: # outliers dataframe
        outliers = data[model.labels_ == -1]
        print(outliers)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
14	15	5.8	4.0	1.2	0.2	
15	16	5.7	4.4	1.5	0.4	
41	42	4.5	2.3	1.3	0.3	
57	58	4.9	2.4	3.3	1.0	
60	61	5.0	2.0	3.5	1.0	
62	63	6.0	2.2	4.0	1.0	
93	94	5.0	2.3	3.3	1.0	
98	99	5.1	2.5	3.0	1.1	
106	107	4.9	2.5	4.5	1.7	
117	118	7.7	3.8	6.7	2.2	
118	119	7.7	2.6	6.9	2.3	
119	120	6.0	2.2	5.0	1.5	
122	123	7.7	2.8	6.7	2.0	
131	132	7.9	3.8	6.4	2.0	
135	136	7.7	3.0	6.1	2.3	

	Species
14	Iris-setosa
15	Iris-setosa
41	Iris-setosa
57	Iris-versicolor
60	Iris-versicolor
62	Iris-versicolor
93	Iris-versicolor
98	Iris-versicolor
106	Iris-virginica
117	Iris-virginica
118	Iris-virginica
119	Iris-virginica
122	Iris-virginica
131	Iris-virginica
135	Iris-virginica

In []: